

- 5 (a)** field strength = potential gradient [- sign not required]
[allow $E = \Delta V/\Delta x$ but not $E = V/d$] **B1 [1]**
- (b)** No field for $x < r$ **B1**
for $x > r$, curve in correct direction, not going to zero **B1**
discontinuity at $x = r$ (vertical line required) **B1 [3]**

- 6 (a) (i)** flux/field in core must be changing **M1**